

When Better Detectors Break Their Monitors: Detector-Quality Dependence and a Null-Detector Diagnostic for Conformal Perception Monitoring

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Abstract

Conformal prediction combined with anytime-valid sequential testing offers a principled way to monitor a deployed perception stack: calibrate a nonconformity score on nominal data, track its miscoverage on the live stream, and alarm when a betting process accumulates enough evidence that the nominal assumption has failed. We set out to apply this recipe to multimodal 3D detection under real adverse weather and to improve its detection delay. We report instead three findings we believe are more useful, and a plain negative result on the original goal.

First, a monitor of this kind can produce entirely plausible numbers while measuring nothing about the detector. We introduce a *null-detector diagnostic* — rerun the pipeline with the detector’s regression head output replaced by zeros — which costs no additional data, training or annotation. On two real adverse-weather datasets it reproduced our own previously reported detection delays *exactly*, to the frame, from a model with no functioning detection head, because a collapsed detector reduces the nonconformity score to a function of the ground-truth labels alone. We recommend this control as standard practice.

Second, split conformal’s exchangeability assumption fails across driving sessions. On the Canadian Adverse Driving Conditions dataset we measure a $7.5\times$ spread in per-drive miscoverage under a single pooled quantile (0.083 to 0.623 against a target of 0.20), with between-session standard deviation as large as the weather effect itself. Mondrian session-conditional calibration repairs this

on a mid-quality checkpoint (effect-to-noise $0.55 \rightarrow 0.85$; worst-case per-session deviation $0.41 \rightarrow 0.19$), but that advantage *reverses* on a converged checkpoint, so the benefit is model-dependent rather than unconditional.

Third, and most consequentially, monitorability is *non-monotone in detector quality*. Across five trained checkpoints spanning a real quality range, the measured weather effect declines monotonically as the detector improves ($0.108 \rightarrow 0.007$ for a phantom-detection score) and becomes statistically undetectable beyond a heatmap separation of roughly 0.41. Two structurally different nonconformity scores — one blind to phantom detections, one dominated by them — agree on this. The practical warning is uncomfortable: *improving your detector can silently disable your monitor*, because a detector that degrades gracefully leaves no error differential to detect, and the monitor keeps emitting plausible numbers throughout.

We are explicit about power: only two of six curve points reach significance with 18 sessions, so the claim rests on the monotone effect-size trend and the loss of detectability, not on a series of significant measurements. On the original objective we are equally plain — at the effect sizes available here reliable detection is not achievable, and we report that with intervals rather than burying it.

Keywords: Conformal prediction, Anytime-valid sequential testing, Perception monitoring, Mondrian conformal prediction, Distribution shift, Autonomous vehicle perception, Adverse weather, Negative controls

1. Introduction

A deployed autonomous-vehicle perception stack is validated under some set of sensing conditions and then asked to operate outside them. The failure that matters operationally is not that the detector is sometimes wrong — it always
5 is — but that it becomes wrong in a way nothing downstream is told about. Conformal prediction offers a distribution-free way to attach calibrated uncertainty to a black-box detector’s outputs, and anytime-valid sequential testing

turns that into a live monitor: calibrate a nonconformity score on nominal data, track its miscoverage $m(t)$ on the incoming stream, and raise an alarm when a betting process accumulates enough evidence against the nominal hypothesis. The construction is attractive because its false-alarm guarantee holds uniformly over all stopping times, so the question “has anything changed yet?” can be asked continuously without a multiple-testing penalty.

We set out to apply that recipe to multimodal 3D detection under real adverse weather, and specifically to improve its detection delay. We did not achieve that, and this paper is largely about why the attempt was more instructive than the goal. Three findings emerged that we think are more useful than a delay improvement would have been, and we state the negative result on the original objective plainly rather than burying it.

A monitor can be entirely fictitious and look fine.. The nonconformity score in this setting is a residual between predicted and ground-truth boxes, evaluated at cells a ground-truth object occupies. If the detector’s regression head collapses — as ours did, under a placeholder training loss whose optimum is a constant — that residual degenerates to a function of the ground-truth labels alone. The monitor then measures the distribution of scene content, not detector error, and because scene content genuinely differs between a dataset’s clear and adverse splits, it produces detection delays that look completely reasonable. We introduce a *null-detector diagnostic* that catches this: rerun the entire pipeline with the regression head’s output replaced by zeros. It requires no additional data, training or annotation. Applied to our own earlier results, it reproduced every reported detection delay *exactly*, to the frame, on both datasets — a monitor with no detector performing identically to one with a detector (Section 4.2). We think this control belongs in any conformal monitoring study, and we foreground it as a contribution in its own right rather than as an internal check.

Exchangeability fails across driving sessions.. Split conformal calibration assumes calibration and test data are exchangeable. Real driving data is collected in sessions, and sessions differ in ways that survive calibration: on the Canadian

Adverse Driving Conditions dataset (CADC) [1] we measure per-drive mean mis-coverage ranging from 0.083 to 0.623 against a target of 0.20 — a $7.5\times$ spread
 40 across drives that are all nominal, all clear-road, several from a single collection date. The between-session standard deviation is as large as the weather effect the monitor exists to detect, so under pooled calibration the monitored quantity is dominated by which session the vehicle is on. Mondrian conformal prediction [2], conditioning calibration on the session, repairs this on one checkpoint
 45 — but, as we report in Section 4.4, that advantage reverses on another, so we present it as model-dependent rather than as a general property.

Monitorability is non-monotone in detector quality.. This is the paper’s central claim and the most counterintuitive. Measuring the weather effect across a series of checkpoints spanning a real quality range, the effect is largest at intermediate
 50 detector quality and declines monotonically as the detector improves, becoming statistically undetectable beyond a heatmap object-versus-empty separation of roughly 0.41 (Section 4.7). A collapsed detector yields only content artifacts; a mid-quality detector yields real, attributable weather signal; a converged detector yields none, because it degrades gracefully enough under snow that little
 55 error differential remains. Two structurally different nonconformity scores — one restricted to ground-truth object cells and therefore blind to spurious detections, one built entirely from activation in empty space — agree on where signal exists, which is stronger evidence than either could provide alone.

The deployment implication is uncomfortable and, we think, worth stating
 60 directly: *improving your detector can silently disable your monitor*. The monitor does not fail loudly when this happens. It continues to emit calibrated-looking quantiles and plausible alarm times, and without a control of the kind described above there is no signal that anything is wrong.

What we do not claim.. We are explicit about statistical power throughout
 65 rather than confining it to a limitations section. The quality curve has six points, of which only two reach significance with 18 sessions; the claim rests on the monotone effect-size trend and the loss of detectability, not on a run of

significant measurements. One of the six — the collapsed checkpoint — is not a commensurable measurement at all, since its score is a demonstrated content artifact, and we mark it as such rather than fitting a trend through it. On the original objective, detection at these effect sizes is not achievable: the best configuration we found controls false alarms where the baseline does not, but detects roughly 22% of onsets, no better than a null detector (Section 4.8). We report that with intervals attached.

Snowy Scenes [3] appears throughout as a negative control. Its real detector shows no weather effect distinguishable from zero, while its null detector shows a large one — the mirror image of CADC — and that contrast is what makes CADC’s mid-quality result credible rather than a diagnostic artifact of our own pipeline.

2. Related Work

2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature concatenation toward BEV-centric architectures that project both modalities into a shared spatial representation before fusing, with attention-based mechanisms used to weight each modality’s contribution per region. A separate line of work has shown that such fusion is not automatically robust: a corrupted or adversarially perturbed single modality can dominate a static or softly-attended fusion decision and collapse detection performance, since nothing in the fusion mechanism verifies that the modalities being combined actually agree geometrically before combining them. The detector this paper monitors adopts a verify-then-fuse design intended to address exactly that failure mode — a Cross-modal Consistency Probe that computes a per-BEV-cell consistency score between camera and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines a disagreeing modality at the grid-cell level, trained under an Adversarial Consistency Training objective. We treat this design here strictly as background infrastructure and describe only as much of its mechanism (Section 3) as this

paper’s own monitoring layer depends on, since the monitor itself is detector-agnostic.

2.2. Conformal Prediction and Anytime-Valid Sequential Testing

100 Split conformal prediction [4, 5] gives distribution-free, finite-sample coverage guarantees for a black-box predictor’s outputs by calibrating a nonconformity score on held-out exchangeable data; the guarantee is a batch one, valid at a single fixed evaluation, and does not by itself say anything about monitoring a live, continuously arriving stream. Sequential testing by betting [6, 7] closes
 105 that gap: framing a hypothesis test as an accumulating wealth process that is a nonnegative supermartingale under the null lets Ville’s inequality bound the false-alarm probability uniformly over all stopping times, giving an anytime-valid guarantee with no fixed monitoring horizon and no post-hoc multiple-testing correction. The direct predecessor for applying this backbone to a live
 110 perception-failure problem is Monroy Muñoz, Verma, and Timans’s work on sequential object-tracking-failure detection [8], which frames tracking failure as an e-process over a bounded tracking-quality metric with the betting rate learned adaptively from the metric’s own history (approximate GRAPA or Scale-Free Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks.
 115 It is a strong, directly applicable baseline, and everywhere in this paper that we compare against a "covariate-blind" betting rule, we mean their rule, reimplemented exactly, not a strawman.

Adjacent recent work conditions conformal guarantees on covariates or system state without adopting the sequential/anytime-valid framing: state-dependent
 120 conformal prediction for perception-based autonomous systems [9] combines conformal error bounds with symbolic reachability analysis but remains a batch procedure; context-aware nonconformity functions for robotic planning and perception [10] learn context-conditioned nonconformity scores, with context drawn from learned task representations rather than a cross-modal sensor-disagreement
 125 signal specifically; and a value-gated modality-refiner approach [11] uses cross-modal agreement to gate fusion, but as a learned neural gate rather than a

statistical test covariate, and targets sentiment/action-recognition/audio-visual tasks rather than autonomous-vehicle perception. None of these combine sequential anytime-valid testing, spatially-resolved multi-object structure with
130 multiplicity control, and a betting rule conditioned on an external cross-modal disagreement signal – the combination this paper’s monitoring layer targets, following the sharpened contribution statement worked out in the ICRA-targeted predecessor to this paper’s monitoring contribution.

2.3. Conditional Validity and Negative Controls

135 Marginal conformal validity holds on average over the calibration distribution and can fail badly within identifiable subgroups. Mondrian conformal prediction [2] addresses this by conditioning calibration on a taxonomy, giving validity within each category rather than marginally; the textbook treatment appears in Vovk, Gammerman and Shafer [4]. The taxonomy is usually a class
140 label. We apply it with the driving session as the category, which is the grouping along which we measure exchangeability to fail in real autonomous-driving data (Section 4.3), and we report a case where the resulting benefit reverses.

We are not aware of prior work that reports a null-detector control for a conformal perception monitor – that is, rerunning the pipeline with the monitored
145 model’s predictions replaced by a constant to establish that the monitor responds to model error rather than to dataset structure. The closest established practice is the use of permutation and label-shuffling baselines in the broader evaluation literature, which serve the same epistemic purpose: distinguishing a method’s measured effect from structure already present in the data. Our
150 contribution here is not the idea of a negative control but the specific, cheap instantiation for conformal monitoring, together with evidence that it changes conclusions on real published-style numbers (Section 4.2).

2.4. Adverse-Weather Datasets for Autonomous Perception

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and
155 Waymo Open Dataset predominantly capture clear-weather conditions, limit-

ing their use for evaluating perception under real distribution shift from adverse weather. Two existing datasets focus specifically on snowy conditions: the Canadian Adverse Driving Conditions dataset (CADC) [1], the first fully snowy dataset built for 3D object detection, and the Winter Adverse Driving dataSet (WADS), the first to add 3D semantic segmentation and individual-snowflake labeling; both, however, are reported to have comparatively low-resolution or unimodal sensing relative to more recent captures. Bijelic et al.’s fog- and rain-focused multimodal fusion work [12] addresses an adjacent adverse-weather regime with a different sensing challenge (fog/rain rather than snow) and is not built around a clear-to-degraded onset transition. Snowy Scenes [3], described fully in Section 4, is a substantially more recent (2026) multimodal dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB, and thermal cameras; the price of that specificity, as we found once real access arrived, is that it contains no clear-weather split at all, which this paper addresses by measuring a real, physically grounded severity ordering across its own weather categories rather than assuming the dataset would hand us the clear/snow contrast originally planned around. This paper’s own contribution to that evaluation landscape is not a new dataset, but the first application, to our knowledge, of an anytime-valid sequential monitor to real (rather than synthetically corrupted) multimodal adverse-weather driving data, replicated across two independent such datasets (Snowy Scenes and CADC) rather than one.

3. Method

3.1. Background: The Monitored Detector

The detector this paper monitors is a camera/LiDAR BEV fusion architecture; we summarize only the components this paper’s monitoring layer directly depends on, since the monitor itself is detector-agnostic and this paper’s own contribution begins with the monitoring layer described in the rest of this section.

Camera and LiDAR streams are each encoded into a shared bird’s-eye-view (BEV) feature grid, $F_{\text{cam}}, F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. A Cross-modal Consistency Probe (CCP) computes, per BEV grid cell (i, j) , a consistency score

$$S(i, j) = \sigma \left(\text{MLP}_S [F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|] \right) \in [0, 1], \quad (1)$$

from the concatenated ($\oplus$) camera and LiDAR feature vectors at that cell and their absolute difference, so that $1 - S(i, j)$ is a cross-modal disagreement signal intended to flag localized geometric or semantic mismatch between the two modalities. A Gated Adaptive Fusion Module (GAFM) uses S to weight each modality’s contribution to the fused BEV representation at the grid-cell level, and an Adversarial Consistency Training (ACT) objective trains the detector, the CCP, and the fusion gate jointly under a loss combining detection accuracy, a CCP contrastive-consistency term, and an attribution-regularization term, with adversarially perturbed camera and LiDAR branches included in the training loop to encourage S to remain informative under perturbation. This paper’s own contribution begins with the next subsection: everything from here on – the conformal calibration, the sequential test, the spatial multiplicity control, and the use (and, as Section 4 reports, the eventual disuse) of S as a monitoring covariate – is this paper’s method, evaluated against real data in Section 4.

Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

3.2. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \dots$, one per LiDAR sweep. At every t , the base 3D detector f_θ described above produces per-object box predictions from the frame’s fused camera/LiDAR BEV features; the monitor itself is detector-agnostic and does not depend on f_θ ’s specific architecture. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object i in frame t , the nonconformity score is the ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually

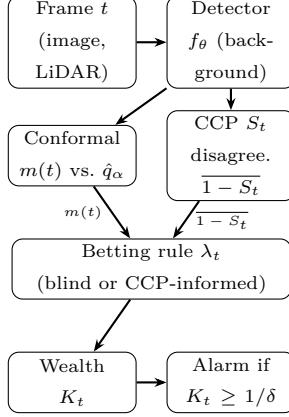


Figure 1: Per-frame monitoring pipeline. The detector f_θ (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score S_t ; everything downstream of it is this paper’s own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

occupies,

$$s_i(t) = \|\hat{B}_i(t) - B_i(t)\|_1,$$

and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal
215 quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set’s non-conformity scores $\{s_i\}_{i=1}^n$, so that a new object is covered iff $s_i(t) \leq \hat{q}_\alpha$. The frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty
frame carries no evidence of miscoverage. The null hypothesis the monitor tests
220 is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$ for all t ; the goal is to detect the first violation – weather-onset degradation – with an anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

3.3. The e -Process and Ville's Inequality

225 The wealth process is

$$K_0 = 1,$$

$$K_t = \prod_{s=1}^t (1 + \lambda_s(m(s) - \alpha)), \quad \lambda_s \in [0, \frac{1}{\alpha}] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably.} \quad (2)$$

We work through why this construction gives an anytime-valid test, rather than only stating the result, since the two steps – the supermartingale property and Ville's inequality – are each short but easy to state imprecisely.

Step 1: K_t is a nonnegative supermartingale under H_0 . Non-negativity requires $1 + \lambda_t(m(t) - \alpha) \geq 0$ for every realizable $m(t) \in [0, 1]$. Since $\lambda_t \geq 0$, the expression is affine and non-decreasing in $m(t)$, so its minimum over $m(t) \in [0, 1]$ occurs at $m(t) = 0$, giving $1 - \lambda_t \alpha$. This is non-negative exactly when $\lambda_t \leq 1/\alpha$ – the bound our implementation enforces on every betting rule (`WealthProcess.lambda_max` in `conformal_monitor/betting.py`), and
 230 the reason the bound is $1/\alpha$ specifically, not the also-plausible-looking $1/(1 - \alpha)$ (which only bounds the opposite endpoint, $m(t) = 1$, and is too loose at the endpoint that actually binds). For the supermartingale property itself, λ_t is chosen $\mathcal{F}_{t-1}$ -measurably – it may depend on the entire history up to and including frame $t - 1$, but not on $m(t)$ itself – so, conditioning on $\mathcal{F}_{t-1}$,

$$\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] = K_{t-1} \cdot \mathbb{E}[1 + \lambda_t(m(t) - \alpha) \mid \mathcal{F}_{t-1}] = K_{t-1} (1 + \lambda_t(\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] - \alpha)),$$

240 using that λ_t and K_{t-1} are both $\mathcal{F}_{t-1}$ -measurable and can be pulled outside the conditional expectation. Under H_0 , $\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$, so the bracketed term is ≤ 1 whenever $\lambda_t \geq 0$ (which it always is, by construction), giving $\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] \leq K_{t-1}$: precisely the supermartingale inequality, holding for *every* choice of predictable, non-negative $\lambda_t \leq 1/\alpha$, which is exactly the freedom the
 245 different betting rules in Sections 3.4–3.5 exploit without ever needing to reprove this step.

Step 2: Ville's inequality bounds the running maximum.. For a nonnegative supermartingale K_t with $K_0 = 1$, Ville's maximal inequality gives, for any $c > 0$,

$$\mathbb{P}_{H_0}\left(\sup_{t \geq 0} K_t \geq c\right) \leq \frac{\mathbb{E}[K_0]}{c} = \frac{1}{c}.$$

250 Setting $c = 1/\delta$ gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$. The alarm rule – reject H_0 the first time $K_t \geq 1/\delta$ – therefore has false-alarm probability at most δ *regardless of the (random) stopping time at which the process is examined*: the guarantee is over the entire trajectory's supremum, not over K_t at one pre-specified t , which is exactly what makes checking continuously, with no fixed monitoring
255 horizon and no correction for how many times the test has been evaluated, statistically valid rather than an uncorrected multiple-testing error. This two-step argument – the supermartingale property from Step 1, Ville's inequality from Step 2 – is the established backbone of testing by betting and is not this paper's contribution; what varies across the rest of this section, and what this
260 paper's own contribution actually depends on, is how the predictable sequence λ_t is chosen.

3.3.1. A Worked Numeric Example

To build intuition before the general betting rules, we trace K_t over five synthetic frames by hand, with $\alpha = 0.2$ (so $\lambda_{\max} = 5$) and a fixed, non-adaptive
265 $\lambda_t = 2$ for simplicity (an actual bettor updates λ_t per Sections 3.4–3.5; holding it fixed here isolates what the recursion itself does). Suppose the observed miscoverage sequence is $m = (0.1, 0.15, 0.6, 0.55, 0.5)$ – nominal for the first two frames, then abruptly degraded from frame 3 onward, mimicking a weather

onset at $t = 3$:

$$\begin{aligned} K_1 &= K_0(1 + 2(0.10 - 0.2)) = 1 \times 0.80 = 0.80, \\ K_2 &= K_1(1 + 2(0.15 - 0.2)) = 0.80 \times 0.90 = 0.72, \\ K_3 &= K_2(1 + 2(0.60 - 0.2)) = 0.72 \times 1.80 = 1.296, \\ K_4 &= K_3(1 + 2(0.55 - 0.2)) = 1.296 \times 1.70 = 2.203, \\ K_5 &= K_4(1 + 2(0.50 - 0.2)) = 2.203 \times 1.60 = 3.525. \end{aligned}$$

270 Two things are worth noticing in this trace. First, wealth *shrinks* while $m(t) < \alpha$ (frames 1–2): a bettor staking $\lambda_t > 0$ on “miscoverage will exceed α ” loses money, by construction, when the pipeline is behaving nominally – this is the price of testing at all, and is what keeps the process a supermartingale rather than an unconditionally growing one. Second, wealth grows multiplicatively, not
275 additively, once $m(t) > \alpha$ consistently (frames 3–5): each factor > 1 compounds against the previous wealth, which is why a real onset that persists for several frames can cross a large threshold $1/\delta$ (e.g. $1/\delta = 20$ for $\delta = 0.05$) well before $t \rightarrow \infty$, even though no single frame’s factor is large by itself. In this toy trace, $K_5 = 3.525$ would already trigger an alarm at $\delta = 0.3$ ($1/\delta \approx 3.33$) but not yet
280 at $\delta = 0.1$ or $\delta = 0.05$ – illustrating concretely why detection delay lengthens as δ shrinks (Table 8): a smaller false-alarm budget requires more accumulated evidence (a higher wealth threshold) before the anytime-valid guarantee permits an alarm.

3.4. Covariate-Blind Baselines

285 Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ itself, reproducing [8] as the baseline to beat. Approximate GRAPA, with $X_t = m(t) - \alpha$, sets

$$\lambda_t = \text{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X_{<t}^2] + \varepsilon}, 0, \lambda_{\max}\right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on

290 the log-wealth gradient, with $g_t = X_t / (1 + \lambda_t X_t)$ and $\eta_t = \lambda_{\max} / \sqrt{1 + \sum_{s \leq t} g_s^2}$,

$$\lambda_{t+1} = \text{clip}(\lambda_t + \eta_t g_t, 0, \lambda_{\max}).$$

Both are exact reimplementations of the adaptive betting rules [8] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

3.5. Covariate-Informed Betting

295 Alongside the two contributions in Sections 3.6 and 4, we also investigated conditioning the betting rate on an external signal available at the same timestep as $m(t)$ but computed from independent evidence: the disagreement output of the monitored detector’s own CCP, $1 - S$ (Equation 1). We describe the mechanism in full because Section 4 reports, in equal detail, why it did not
300 deliver the advantage motivating it – an outcome worth understanding precisely rather than skipping past. Our covariate-informed bettor wraps either covariate-blind base bettor and scales its stake by the frame’s mean disagreement,

$$\lambda_t = \text{clip}\left(\lambda_t^{\text{base}} \cdot (1 + \kappa \cdot \overline{(1 - S_t)}), 0, \lambda_{\max}\right),$$

where $\kappa \geq 0$ gates the covariate’s influence ($\kappa = 0$ recovers the base bettor exactly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended
305 mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as the lagging one the covariate-blind rules are stuck reacting to. Section 4 reports what we found when we tested that mechanism against real data: two distinct
310 failures, neither in the betting-rule mathematics itself, that between them mean this specific mechanism does not yet deliver the advantage it is designed to.

3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone wrong" – not useful to a downstream planner that needs to know *which* region

315 of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$
 layout of cell-clusters (average-pooling per-cell miscoverage and disagreement
 down to this resolution) and run one independent sequential tester per cluster,
 each with its own wealth process and its own instance of whichever bettor is
 under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires
 320 multiplicity control or the overall false-alarm rate inflates with n ; we support
 two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell
 alarms, marks it permanently flagged – valid under arbitrary dependence but
 conservative, and the sticky-once-alarmed semantics is what we use for the per-
 cell detection-delay evaluation regardless of which regime produced the alarm.
 325 The e-BH alternative treats each cell's current wealth $K_t^{(i)}$ as an e-value: for
 a nonnegative supermartingale started at $K_0 = 1$, $\mathbb{E}_{H_0}[K_t^{(i)}] \leq \mathbb{E}_{H_0}[K_0^{(i)}] = 1$
 at any fixed t (immediate from Section 3.3's Step 1, since the supermartingale
 inequality applied recursively from $t = 0$ gives $\mathbb{E}[K_t] \leq \mathbb{E}[K_{t-1}] \leq \dots \leq K_0 = 1$)
 – exactly the definition of an e-value under $H_0^{(i)}$: a nonnegative random variable
 330 with expectation at most 1. Given $n = n_h n_w$ such e-values at a fixed t , one per
 cell, e-BH [13] rejects the set

$$R_t = \left\{ i : K_t^{(i)} \geq \frac{n}{k^* q} \right\}, \quad k^* = \max \left\{ k \in \{0, \dots, n\} : e_{(k)} \geq \frac{n}{k q} \right\},$$

where $e_{(1)} \geq e_{(2)} \geq \dots \geq e_{(n)}$ is the descending order statistic of the n cur-
 rent wealths, and $k^* = 0$ gives $R_t = \emptyset$. The rule is a direct e-value analogue
 of Benjamini-Hochberg: it thresholds the k -th largest e-value against a level
 335 that grows with k (more permissively for larger candidate rejection sets), then
 commits to the *largest* such k – the same "largest crossing index" structure as
 classical BH's p -value threshold, but with the inequality direction reversed be-
 cause e-values are large, not small, under evidence against the null. Its false
 discovery rate guarantee, $\mathbb{E}[\text{FDR}_t] \leq q$ at every fixed t , holds under *arbitrary*
 340 *dependence* among the n e-values – unlike classical BH on p -values, which re-
 quires independence or specific positive-dependence structures (PRDS) to con-
 trol FDR without an extra $\log n$ correction factor – because the e-value defi-
 nition ($\mathbb{E}[e_i] \leq 1$ under $H_0^{(i)}$) is closed under averaging regardless of the joint

dependence structure of the e_i , which is precisely the guarantee needed here:
 345 spatially adjacent BEV cells’ miscoverage and disagreement signals are driven
 by overlapping physical causes (a single snow squall affects many neighboring
 cells at once) and are certainly not independent. Unlike Bonferroni’s sticky
 once-alarmed flags, e-BH is applied fresh at every timestep from each cell’s cur-
 rent wealth, so it produces a live, non-sticky map that can un-flag a cell once its
 350 local evidence subsides – appropriate for a spatial map meant to reflect *current*,
 not historical, trustworthiness. Algorithm 1 summarizes the per-frame update.

3.6.1. Betting-Rule Comparison: Theoretical Properties and Tradeoffs

Table 1 summarizes the three betting rules evaluated in this paper, all valid
 choices of the predictable λ_t in Equation 2 (so all three inherit Section 3.3’s
 355 anytime-valid guarantee unconditionally; what differs is statistical power, not
 validity). AGRAPA is a plug-in estimator of the (asymptotically) growth-
 rate-optimal constant bet, tracking the running first and second moments of
 $X_t = m(t) - \alpha$ and converging quickly when the true miscoverage rate is close
 to stationary, but reacting relatively slowly to an abrupt regime change, since a
 360 single new observation only nudges a running average computed over all prior
 history. SF-OGD instead takes an online-convex-optimization view, performing
 AdaGrad-style gradient ascent directly on (an approximation to) the log-wealth
 objective with a decaying step size; this makes it more reactive to recent observa-
 tions by construction (the step size η_t shrinks as $1/\sqrt{t}$, but each step still moves
 365 λ_t in the direction the *most recent* gradient indicates), at the cost of higher
 variance in λ_t itself, particularly early in a stream before $\sum_{s \leq t} g_s^2$ has accu-
 mulated enough to stabilize η_t . Both are exact reimplementations of the rules
 [8] uses, run here against a different sensing modality and detection task than
 the 2D visual tracking they were originally evaluated on. The CCP-informed
 370 variant wraps either base rule multiplicatively rather than replacing it, so it
 inherits whichever base rule’s reactivity/stability tradeoff was chosen, and adds
 a single scalar gain κ controlling how strongly the covariate perturbs the base
 bet; $\kappa = 0$ recovers the base rule exactly, which is why Section 4 can report the

Table 1: Betting rules compared in this paper. All three are valid choices of λ_t in Equation 2 and inherit the same anytime-valid guarantee unconditionally (Section 3.3); they differ in statistical power, not validity.

Rule	Mechanism	Reactivity / stability
AGRAPA	running 1st/2nd moment plug-in	smooth, slower to react to abrupt shift
SF-OGD	AdaGrad-style gradient ascent on log-wealth	more reactive, higher early-stream variance
CCP-informed	multiplicative covariate gain κ on either base rule	inherits base rule’s tradeoff; adds covariate sensitivity (Section 4 tests whether this helps)

covariate-informed and covariate-blind results as a controlled comparison (same
375 base rule, same wealth-process mechanics, differing only in whether $\overline{1 - S_t}$ enters
the stake) rather than two unrelated procedures.

3.7. Session-Conditional (Mondrian) Calibration

Split conformal calibration pools nonconformity scores and produces one
quantile, valid on average over the calibration distribution. Real driving data
380 arrives in sessions that differ persistently in weather, lighting, traffic and route,
and Section 4.3 measures the resulting per-session violation directly. Mondrian
conformal prediction [2, 4] conditions calibration on a taxonomy, giving validity
within each category; here the category is the driving session.

Algorithm 1 Per-frame spatial monitoring update

Require: frame t ; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w) ; correction mode

- 1: compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
- 2: compute per-cell CCP disagreement $\overline{(1 - S)}^{(i)}(t)$
- 3: **for** each cluster $i = 1, \dots, n_h n_w$ **do**
- 4: $\lambda_t^{(i)} \leftarrow \text{bettor.NEXT_LAMBDA}(\overline{(1 - S)}^{(i)}(t))$
- 5: $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot (1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha))$
- 6: $\text{bettor.UPDATE}(m^{(i)}(t), \overline{(1 - S)}^{(i)}(t))$
- 7: **end for**
- 8: **if** correction = Bonferroni **then**
- 9: flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
- 10: **else** ▷ e-BH
- 11: flag the top- k^* cells by $K_t^{(i)}$, $k^* = \text{largest } k \text{ with } K_{(k)} \geq (n_h n_w) / (k \delta)$
- 12: **end if**
- 13: **return** flagged-cell map (untrustworthy regions at time t)

Mondrian requires calibration data in every category it is queried about, which forces a within-session split: each session calibrates on its own leading frames and is monitored on the remainder. This is the deployment-faithful arrangement — a vehicle calibrates on the early known-nominal portion of the drive it is currently on, rather than assuming a quantile estimated on previous drives transfers — and degraded frames are scored against the quantile of the session the stream began in, which is what a deployment does when conditions change mid-drive. Because calibration and monitored frames then sit adjacent in time within a correlated stream, we insert a temporal gap between them and sweep its size; an effect inflated by adjacency shrinks as the gap grows.

3.8. A Phantom-Detection Nonconformity Score

The score of Section 3.2 is a box-regression residual restricted to cells a ground-truth object occupies, and is therefore structurally blind to spurious detections in empty space. Snow’s dominant effect on LiDAR is precisely spurious near-range returns, so we add a second score family: the predicted peak class activation at cells containing no ground-truth object, calibrated by the

400 same split-conformal procedure at level α . The frame-level rate is the convex combination

$$m(t) = (1 - w) m_{\text{loc}}(t) + w m_{\text{phantom}}(t), \quad (3)$$

with $w \in [0, 1]$. Each component is calibrated at level α and so has expectation at most α under the nominal distribution; a convex combination of two such quantities inherits the bound, which is exactly the H_0 condition the wealth process of Section 3.3 requires. No new theory is needed: the conformal guarantee
405 is agnostic to the choice of score function. Setting $w = 0$ recovers the original score exactly.

3.9. Block Aggregation

Averaging m over blocks of k consecutive frames before the betting step
410 trades temporal resolution for reduced variance in the quantity the bettor consumes. Under independent frames the reduction would be $\sqrt{k}$; driving frames are strongly correlated, and Section 4.5 measures the realised gain. We keep one trajectory entry per frame regardless of k , so detection delays remain in frame units and are comparable across block sizes and against the unblocked
415 baseline.

4. Experiments

Every number in this section is produced by a script committed alongside the manuscript and written to a JSON artifact named in the corresponding caption, so each table can be regenerated and checked against its source.

4.1. Setup

We evaluate on two real adverse-weather driving datasets. CADC [1] provides 18 nominal (**bare-road**) driving sessions and 14 snow-covered sessions, with the road-condition label taken from the dataset’s own per-drive metadata rather than inferred. Snowy Scenes [3] provides real snowy driving across three
425 weather categories; since it contains no clear-weather split, its nominal/degraded

contrast is constructed across categories using a measured per-point snow-class fraction, and it serves here as a negative control rather than as a second positive result.

The monitored detector is a camera/LiDAR fusion architecture producing a
 430 per-cell class heatmap and box regression on a 128×128 bird’s-eye-view grid. The nonconformity score is the ℓ_1 box-regression residual at cells a ground-truth object occupies; a second score, introduced in Section 4.6, is built from predicted activation at cells that contain no ground-truth object. All monitoring uses $\alpha = 0.2$.

435 *Evaluation harness..* Onset streams splice a nominal prefix with a degraded suffix at a known onset index. Frames within a stream are sampled without replacement from the relevant pool under a per-replicate seed, and every arm of every comparison reuses one shared seed list, so methods are compared on identical frame draws replicate by replicate. This matters: an earlier version of
 440 this harness constructed streams by modular arithmetic with no sampling, so all replicates of a condition were the identical stream and every reported statistic was a single sample measured n times. The signature was visible in the outputs — false-alarm rates landing on exactly 0.00 or 1.00 and censoring on exactly 0/5 or 5/5 — and a regression test now asserts that replicates differ.

445 *Reporting conventions..* Detection-delay comparisons are gated on false-alarm control: a monitor whose empirical false-alarm rate exceeds δ has no valid delay, because it can appear fast by alarming indiscriminately, including before the onset exists. Negative delays in the tables below denote exactly that pathology. Confidence intervals are percentile bootstrap; where sessions are the unit of
 450 analysis they are resampled over sessions rather than frames, since frames within a drive are strongly correlated and frame-level resampling understates intervals.

4.2. The Null-Detector Diagnostic

The diagnostic reruns the complete pipeline — calibration included — with the detector’s box-regression output replaced by zeros, so the nonconformity

Table 2: Null-detector diagnostic on the pre-fix checkpoints. Identical protocol; the sole difference is whether the box head’s output is real or identically zero. Delays are in frames. Artifacts: `zeros_baseline_cadc.json`, `zeros_baseline_snowy.json`.

Dataset	Arm	$\delta = 0.30$	$\delta = 0.10$	$\delta = 0.05$
CADC	real detector	−5.0 (FA 1.00)	−2.0 (FA 1.00)	— (FA 1.00)
	null detector	6.0 (FA 0.00)	8.0 (FA 0.00)	9.0 (FA 0.00)
Snowy Scenes	real detector	— (FA 1.00)	— (FA 0.00)	— (FA 0.00)
	null detector	3.0 (FA 0.00)	10.0 (FA 0.00)	— (FA 0.00)

455 score reduces to $\|B_{\text{target}}\|_1$ and depends only on ground-truth scene content. If a monitor’s behaviour is driven by detector error, zeroing the detector must change it.

Table 2 applies this to the checkpoints that produced our earlier results. On both datasets the null detector reproduces the real detector’s reported detection 460 delays exactly, at every false-alarm budget. On CADC the null monitor is in fact *better behaved* than the real one, holding a 0.00 false-alarm rate where the real monitor’s exceeds every budget.

The null delays of 6/8/9 on CADC and 3/10 on Snowy Scenes are, to the frame, the numbers we had previously reported as real detection results. The 465 explanation is direct: those checkpoints had collapsed under a placeholder detection loss, so they *were* null detectors. A separate measurement makes this exact rather than inferred — on the collapsed checkpoint the nominal-to-degraded jump in m is -0.21573659 , matching the zeros branch to eight decimal places, with an artifact-subtracted residual of exactly 0.00000 (`quality_effect_curve_cadc.json`).

470 We draw a methodological conclusion beyond our own case. A conformal monitor built on a degenerate score does not fail conspicuously; it reports calibrated quantiles and plausible alarm times throughout. The null-detector control costs one extra evaluation pass and distinguishes the two cases unambiguously, and we suggest it be reported as standard.

Table 3: Per-drive mean frame miscoverage under one pooled quantile ($\hat{q} = 1.935$, $\alpha = 0.20$). All **bare** drives are nominal clear-road driving. Artifact: `per_drive_miscoverage_cadc.json`.

Group	n drives	mean m (sd)	range
calibration drives	9	0.252 (0.140)	0.133 – 0.623
nominal drives	9	0.182 (0.061)	0.083 – 0.284
degraded drives	14	0.238 (0.057)	0.128 – 0.329

4.3. Cross-Session Exchangeability Failure

Table 3 reports per-drive mean miscoverage on CADC under a single pooled quantile, using a working (non-collapsed) detector.

Marginal validity holds on average while failing per session: four of nine nominal drives sit above α , one calibration drive reaches 0.623, and the spread across all **bare** drives is $7.5\times$. The comparison that matters is between the weather effect and the between-session noise: the degraded-minus-nominal difference is $+0.057$ against a between-drive standard deviation of 0.061, a ratio below one, and six of nine nominal drives fall inside the degraded drives’ range. A monitor calibrated marginally across sessions is therefore substantially detecting session identity.

4.4. Session-Conditional Calibration, and Where It Reverses

Mondrian calibration conditions the quantile on the session. This requires a within-session split — each drive calibrates on its own leading frames and is monitored on the remainder — because under the whole-drive split a monitored drive has no calibration frames and no per-session quantile exists for it. That arrangement is deployment-faithful (a vehicle calibrates on the early known-nominal portion of the drive it is on) but it places calibration and monitored frames adjacent in time, and adjacent driving frames are correlated. We therefore insert a temporal gap and sweep it; an effect inflated by adjacency would shrink as the gap grows.

On the 5-epoch checkpoint Mondrian improves effect-to-noise by $1.54\times$ and

Table 4: Mondrian versus marginal calibration, matched on sessions, frames, gap and cached scores, differing only in which quantile scores each session. Left: 5-epoch checkpoint. Right: converged (20-epoch) checkpoint. Artifacts: `mondrian_gap_sweep_cadc.json`, `mondrian_gap_sweep_cadc_converged.json`.

gap	effect-to-noise		worst $ m - \alpha $	
	marginal	Mondrian	marginal	Mondrian
<i>5-epoch checkpoint</i>				
0	0.551	0.849	0.412	0.187
10	0.528	0.848	0.427	0.190
20	0.558	0.829	0.480	0.200
<i>converged checkpoint</i>				
0	-0.033	-0.025	0.187	0.344
10	-0.104	-0.079	0.221	0.363
20	0.096	0.003	0.210	0.353

more than halves the worst-case per-session deviation, and the effect is stable across gaps (+0.074 at gap 0, +0.087 at gap 20), so adjacency is not inflating it. That worst-case improvement is the mechanism by which false-alarm control
500 returns, since a 0.41 deviation under pooled calibration is what drives clear-stream alarms.

The converged checkpoint reverses the worst-case ordering: Mondrian is roughly $1.7\times$ *worse* than marginal. We do not have an explanation for this and do not offer one. It means the benefit of session-conditional calibration here is a
505 property of the model–data combination rather than an unconditional improvement, and we state it that way: any claim of the form “Mondrian improves worst-case per-session coverage” is contradicted by our own second measurement.

Table 5: Per-block discriminability $d(k) = [\bar{m}_{\text{deg}} - \bar{m}_{\text{nom}}]/\text{sd}(m_{\text{nom}})$ over blocks of k frames, 5-epoch checkpoint, Mondrian calibration. Artifact: `block_sweep_cadc.json`.

k	block noise	$d(k)$	gain vs $k = 1$	$\sqrt{k}$ predicts
1	0.1452	0.454	$1.00\times$	$1.00\times$
2	0.1260	0.523	$1.15\times$	$1.41\times$
4	0.1118	0.588	$1.30\times$	$2.00\times$
8	0.0950	0.646	$1.42\times$	$2.83\times$

4.5. Block Aggregation and the Temporal-Correlation Penalty

510 Averaging m over blocks of k frames before the betting step cannot change the session-level statistic — averaging within a session leaves the session mean unchanged, so the session-level effect and between-session standard deviation are exactly invariant to k , which we verify to six decimal places as a correctness check on the implementation. What blocking reduces is the within-session per-
515 block variance, the noise a sequential test consumes when judging whether the current stretch of frames looks degraded.

Blocking delivers roughly half the gain independent sampling would predict — $1.42\times$ at $k = 8$ against $2.83\times$ — which quantifies the temporal-correlation penalty for 10 Hz driving frames. At $k = 8$ the effect itself begins to erode
520 $(+0.0659 \rightarrow +0.0614)$ as blocks span genuinely different scene content, so the gain turns over; we use $k = 4$. The null detector’s discriminability stays flat and low across all k (0.128, 0.133, 0.128, 0.100), confirming that blocking does not manufacture discriminability where there is no signal.

4.6. A Phantom-Detection Score

525 The localization score is restricted to ground-truth object cells and is therefore structurally blind to spurious detections in empty space — which is snow’s documented physical effect on LiDAR. We add a second score family: predicted peak activation at cells containing no ground-truth object, calibrated by the same split-conformal procedure at level α , with the frame rate a convex com-

Table 6: Phantom-score effect on the 5-epoch checkpoint, session-paired, session-level bootstrap over 18 sessions. All entries significant (CI excludes zero) at every gap. Artifact: `phantom_effect_5epoch.json`.

w	effect (gap 0)	95% CI	effect (gap 20)	best eff/noise
0.00	+0.0647	[+0.0135, +0.1076]	+0.0772	0.740
0.25	+0.0734	[+0.0338, +0.1102]	+0.0844	0.851
0.50	+0.0821	[+0.0446, +0.1227]	+0.0917	0.862
0.75	+0.0908	[+0.0411, +0.1427]	+0.0990	0.812
1.00	+0.0995	[+0.0304, +0.1714]	+0.1062	0.746

530 bination $m = (1 - w)m_{\text{loc}} + w m_{\text{phantom}}$. Each component has expectation at most α under the nominal distribution, so the combination does too, and $w = 0$ recovers the original score exactly.

The effect grows monotonically with phantom weight, 54% larger at $w = 1$ than at $w = 0$, and every cell survives the gap control. Mixing beats either end-
535 point on effect-to-noise (0.862 at $w = 0.5$), so the two score families are partly complementary rather than redundant: phantom activations carry weather information the localization score cannot see. On the converged checkpoint, by contrast, no weight at any gap produces an effect whose interval excludes zero (`phantom_effect_converged.json`).

540 We flag a limitation of this arm specifically. The null-detector control used elsewhere zeroes the box head, which leaves the heatmap untouched and is therefore not a null for a score that reads the heatmap. Zeroing all heads makes every phantom score identical and the component degenerate — a floor by construction rather than an informative comparison. For $w > 0$ the gap
545 sweep and the confidence intervals carry the evidential weight, and we do not present the all-heads-null row as a control.

4.7. Monitorability Versus Detector Quality

Table 7 measures both score families across six checkpoints on one protocol. The x -axis is heatmap object-versus-empty separation, not epoch count or

Table 7: Weather effect versus detector quality, gap 10, session-level bootstrap over 18 sessions. * marks a 95% CI excluding zero. The collapsed checkpoint is **not a commensurable measurement** — its score is a demonstrated content artifact (Section 4.2) rather than detector error — and is shown as an artifact reference, excluded from the trend. Artifact: `quality_effect_curve_cadc.json`.

Checkpoint	separation	localization effect	phantom effect
<i>artifact reference, not part of the trend</i>			
collapsed	−0.0002	+0.0341	+0.1562
<i>trained checkpoints</i>			
epoch 5	+0.2500	+0.0660*	+0.1077*
epoch 9	+0.3796	+0.0490*	+0.0205
epoch 13	+0.4148	+0.0116	+0.0191
epoch 16	+0.4488	+0.0353	+0.0161
epoch 20	+0.4844	−0.0126	+0.0074

550 training loss: between epochs 12 and 20 the loss continued to fall ($5.52 \rightarrow 4.87$) while separation plateaued, so an epoch axis would suggest the detector kept improving in the dimension the monitor consumes when it had stopped.

Power, stated before the interpretation.. Only two of the six rows reach significance, and the session-level bootstrap has 18 independent units, so intervals
555 are wide. The claim below rests on the monotone decline in effect size across the five trained checkpoints and on the loss of detectability beyond separation ≈ 0.41 , *not* on a sequence of significant measurements. A reader who requires significance at every point should regard this as suggestive.

With that stated: the phantom effect declines monotonically across all five
560 trained checkpoints, $+0.1077 \rightarrow +0.0074$, as separation rises from 0.25 to 0.48. The localization effect broadly tracks it, from +0.0660 to −0.0126, though not strictly monotonically — epoch 16 (+0.0353) sits above epoch 13 (+0.0116), and neither is significant. The two scores agree on significance at five of six points; they disagree at epoch 9, where localization is significant and phantom

Table 8: CADC operating curve, 36 replicates, shared seeds across arms. “det.” is the number of replicates on which the monitor alarmed after onset. Artifacts: `window_sweep_cadc.json`, `final_operating_curve_cadc.json`.

Arm	δ	FA	FA 95% CI	det.	delay
baseline (marginal, $k=1$)	0.30	0.14	[0.03, 0.25]	5/36	−9.4
	0.10	0.11	[0.03, 0.22]	4/36	−7.5 (FA > δ)
	0.05	0.11	[0.03, 0.22]	4/36	−5.8 (FA > δ)
ours (Mondrian, $k=4$)	0.30	0.17	[0.06, 0.28]	11/36	12.8
	0.10	0.03	[0.00, 0.08]	8/36	18.0
	0.05	0.00	[0.00, 0.00]	6/36	19.7
null (Mondrian, $k=4$)	0.30	0.19	[0.08, 0.33]	8/36	0.5
	0.10	0.17	[0.06, 0.28]	8/36	9.0 (FA > δ)
	0.05	0.17	[0.06, 0.31]	7/36	12.1 (FA > δ)

565 is not. We report the disagreement rather than smoothing it: at that point the two score families give different answers about whether signal remains.

The collapsed checkpoint’s apparent phantom effect of +0.1562 is the largest in the table and is not detector signal at all. Including it in a fitted trend would be a category error, which is why it is separated above.

570 4.8. Operating Curve

Table 8 gives the end-to-end result on the 5-epoch checkpoint — the configuration where signal exists — at the best window length found (40 frames, onset at 40%).

Two things hold. The baseline is invalid at $\delta \leq 0.10$ and its delays are
575 *negative* — it alarms before the onset exists, and does so progressively earlier as the window lengthens (−3.4 at 20 frames, −9.4 at 40, −17.3 at 80), which is the signature of accumulating on session identity. Our arm controls false alarms at every budget where the baseline does not, and the null detector is invalid at $\delta \leq 0.10$, so where our arm is valid and detecting, the detection is attributable.

580 One thing does not hold, and it is the original objective. Detection rate is 31%, 22% and 17% — and on detection rate alone the null detector matches it (8/36 versus 8/36 at $\delta = 0.10$). Longer windows do not fix this: across 20/40/60/80 frames detection rises from 8% to 22% and then plateaus, while false-alarm control breaks at 80 frames as the extra frames give the clear streams
585 more opportunity to cross the threshold. We therefore make no performance claim. The monitor we can build is *valid* where the standard one is not; it is not *reliable*.

4.9. Snowy Scenes as Negative Control

On Snowy Scenes the real detector’s weather effect is -0.0085 with a 95% CI
590 of $[-0.0719, +0.0553]$ — centred on zero — while the null detector’s is -0.1315 , $[-0.1917, -0.0726]$, clearly non-zero (`effect_exists_snowy.json`). The only signal present is a content artifact pointing the wrong way, which is why the null monitor “detected” at 3 and 10 frames there while the real one was censored throughout.

595 This is the mirror image of CADC, where the real effect is present and the null’s is not, and the contrast is what makes the CADC result credible: the same diagnostic applied to two datasets returns opposite verdicts, so it is discriminating between them rather than reporting a property of our pipeline.

5. Discussion

600 5.1. Why a Fictitious Monitor Looks Healthy

The most uncomfortable result in this paper is that our own reported detection delays were reproduced exactly, to the frame, by a detector that outputs nothing. It is worth being precise about why, because the mechanism generalizes past our particular bug.

605 The nonconformity score is a residual, $\|B_{\text{pred}} - B_{\text{target}}\|_1$, evaluated where a ground-truth object sits. When the prediction term vanishes the score becomes $\|B_{\text{target}}\|_1$: a function of the labels alone. Nothing downstream notices.

Split conformal calibration still returns a well-defined quantile, because it is a quantile of *some* exchangeable score distribution. The betting process still
610 accumulates wealth, because $m(t)$ still moves. The alarm still fires at a plausible time, because scene content genuinely differs between a dataset’s nominal and adverse splits — on CADC the mean ground-truth object count per frame roughly doubles. Every component behaves exactly as specified. The monitor is measuring the dataset’s annotation statistics and reporting them as perception
615 degradation.

There is an inversion here worth naming. On CADC the null monitor was *better behaved* than the real one: false-alarm rate 0.00 against 1.00. The reason is that scene content is relatively homogeneous across nominal drives, so a content-derived score transfers cleanly between sessions, whereas genuine detector error
620 is heterogeneous across sessions and does not. The artifact was better behaved than the truth. This inverts the usual heuristic that cleaner numbers indicate a better method, and it is precisely why the failure is hard to notice from the outputs alone.

The practical recommendation is cheap. One extra evaluation pass with the
625 prediction head zeroed, reported as a row in the results table, separates “our monitor detects degradation” from “our monitor detects the dataset”. We would not have found our own error without it.

5.2. Sessions Are Not Exchangeable, and What That Costs

Conformal prediction’s guarantee is conditional on exchangeability, and driving data violates it in a specific, structured way: it arrives in sessions, and
630 sessions differ in weather, lighting, traffic density and route in ways that persist across every frame within them. Our measurement puts a number on the cost — a $7.5\times$ spread in per-drive miscoverage under a pooled quantile, with between-session standard deviation (0.061) as large as the weather effect being
635 sought (0.057).

The consequence is not a small loss of efficiency. It is that the monitored quantity is dominated by which session the vehicle is on, so the monitor detects

session identity. That is visible directly in the operating curve: the marginal baseline’s detection delays are negative and become more negative as the window
640 lengthens, because a longer window simply supplies more nominal frames from an unlucky session to accumulate against.

Mondrian calibration is the natural repair and it works on one checkpoint. We report the reversal on the other rather than omitting it, and we deliberately do not explain it, because we cannot. What we can say is that the reversal
645 makes an unconditional recommendation unsafe: whether session-conditional calibration improves worst-case coverage appears to depend on the model being calibrated, and a practitioner should measure it on their own model rather than assume it. We would rather publish a qualified result than a clean one we cannot support at a second measurement.

650 5.3. Monitorability Is Non-Monotone in Detector Quality

The central finding is that the monitorable weather effect is largest at intermediate detector quality and declines as the detector improves, becoming undetectable at our best checkpoint.

The mechanism we propose is straightforward, though we test it only indirectly. A conformal monitor of this design detects a *differential* — the gap
655 between nominal-condition error and degraded-condition error. A collapsed detector has no error differential because it has no error signal at all, only content. A converged detector has little error differential because it handles snow nearly as well as clear conditions: that is what robustness means. The differential is
660 largest for a detector good enough to have real, structured error but not good enough to be robust to the condition being monitored.

If this is right, it is an awkward property for the field. The better-engineered the perception stack, the less there is for a degradation monitor of this design to find, and the monitor’s failure is silent — it continues producing calibrated
665 quantiles and plausible alarm times. A team that improves its detector and keeps its monitor may reasonably believe both are working.

Three qualifications constrain how far this should be taken. First, power:

two of six points significant, 18 sessions, wide intervals; the claim rests on the trend and on the loss of detectability, not on a run of significant results. Second, the two scores disagree at epoch 9, and we do not know which is right there. Third, we vary detector quality along one axis, on one architecture and one dataset; whether the same shape appears for other detectors or other conditions is untested. The finding is suggestive and mechanistically plausible, not established.

It also admits a more optimistic reading we cannot presently distinguish from ours: perhaps a converged detector’s residual degradation is real but concentrated in rarer, harder cases that a frame-averaged miscoverage rate dilutes away. That would make the problem one of *score design* rather than of monitorability as such. Our phantom-score result is weak evidence against the simplest version of that story — a score built entirely from a different failure mode also finds nothing at convergence — but it is not decisive, and a score targeting rare severe errors specifically is the experiment we would run next.

5.4. What the Score Families Tell Us

Adding a phantom-detection component was the only intervention in this project that changed *what is measured* rather than how we bet on it, and it is the only one that enlarged the effect: +54% at $w = 1$ relative to localization alone, with mixing at $w = 0.5$ giving the best effect-to-noise. That the two families are complementary rather than redundant is a genuine methodological result: spurious activation in empty space carries weather information that a matched-cell residual cannot see, which is what one would expect given that snow’s dominant LiDAR effect is spurious near-range returns.

It did not clear the detection bar. Effect-to-noise 0.862 against the 0.849 obtainable without it is within noise of no improvement on that metric, even though the raw effect is substantially larger — the between-session variance grows alongside the effect. We report both numbers rather than the flattering one.

5.5. Limitations

Beyond the power constraints already stated: our evaluation is a single detector architecture on two datasets, with 18 nominal sessions on the dataset carrying the positive result. Training used chunked resumption, so optimizer moment estimates re-warm at each resume — a small deviation from an uninterrupted run, stated because it is a real difference. The within-session split used for Mondrian places calibration and monitored frames in the same drive; we control for adjacency with a temporal-gap sweep and the effect is stable across it, but the split still differs from the whole-drive protocol used elsewhere in this work, and we flag that rather than leaving a reader to find the inconsistency. Finally, the degraded regime on CADC is a different collection date from the nominal one, so weather is confounded with date; the null-detector control bounds how much of the effect that could explain, but does not eliminate the confound.

6. Conclusion

We set out to reduce the detection delay of a conformal, anytime-valid monitor for multimodal 3D perception under real adverse weather. We did not succeed at that, and the reasons turned out to be more informative than the target.

Three results stand. First, a conformal monitor built on a degenerate non-conformity score produces calibrated quantiles, plausible alarm times and clean-looking operating curves while measuring nothing about the detector; the *null-detector diagnostic* — one extra evaluation pass with the prediction head zeroed — exposes this, and on our own earlier results it reproduced every reported detection delay exactly, to the frame, on two datasets. Second, split conformal’s exchangeability assumption fails across driving sessions, with a $7.5\times$ spread in per-drive miscoverage under a pooled quantile and between-session variation as large as the weather effect itself; Mondrian session-conditional calibration repairs this on one checkpoint, though the advantage reverses on another, so

we report it as model-dependent. Third, monitorability is non-monotone in detector quality: the measurable weather effect peaks at intermediate quality and declines to undetectability as the detector converges, with two structurally different nonconformity scores agreeing on the trend.

730 On the original objective we are unambiguous. The best configuration we found controls false alarms where the standard approach does not — the standard approach alarms *before* onset, accumulating on session identity — but detects roughly 22% of onsets, which on detection rate alone a null detector matches. Longer windows raise detection to a plateau and then break false-
735 alarm control. At the effect sizes available here, reliable detection is not achievable, and no combination of calibration scheme, betting rule, block size or window length that we tried changes that. The monitor we can build is valid; it is not reliable.

Four directions follow. The most direct concerns score design: if a converged
740 detector’s residual degradation is real but concentrated in rare, severe failures, a frame-averaged miscoverage rate dilutes it away, and a score targeting the tail rather than the mean might recover what we could not. Our phantom-score result argues weakly against the simplest version of this, but does not settle it. Second, the non-monotonicity should be tested on another architecture and
745 another degradation mode before it is treated as general; we establish it on one detector, one dataset and one axis of quality variation. Third, the Mondrian reversal wants an explanation, since a calibration scheme whose benefit depends on the quality of the model being calibrated is a caveat worth understanding rather than merely recording. Fourth, the spatial construction — per-cell e-
750 processes under multiplicity control — remains untested against ground-truth localized degradation, because neither dataset annotates where degradation occurs.

We would emphasise one practical point over the rest. Throughout this work the failures were silent: a collapsed detector, a degenerate evaluation harness
755 whose replicates were all identical, a monitor detecting annotation statistics, a calibration scheme failing per session while valid on average. In every case

the outputs looked reasonable, and in every case what exposed the problem was a control — zeroing the detector, asserting replicates differ, measuring per session rather than in aggregate, inserting a temporal gap. For monitoring systems specifically, whose entire purpose is to be trusted when nothing else is watching, we think the controls deserve as much attention in reporting as the headline numbers.

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